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COI Name (Institution):	Willson, Richard C (University of Houston)		
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Task Description:	The project has three components. First, we are seeking to develop appropriate microbial monitoring technology that consists of a routine component and a specialty component. As currently envisioned, routine monitoring will initially be based on homogeneous solution assays using molecular beacons that look for the presence of specific genera and species as indicator organisms. This would be supplemented by, and perhaps eventually replaced by, a mass spectrometry approach in which one would obtain a profile of the microbial population in a sample. Finally, a sophisticated general bacterial identification system relying on array technology will be developed. This system would likely only be used in the event of a serious problem. Second, we will conduct experiments with the objective of understanding how a low gravity/high background radiation environment affects bacteria over extended time periods. Third, we will seek to extend the core monitoring technology to detection of microbial escape from an ALS system. The objective is twofold: (1) To develop general and specific microbial monitoring systems that utilize technologies that can		

	be readily implemented in spacecraft. The most useful technologies will be those that are easy to implement or utilize multiple use instrumentation that is likely to be available on future missions. (2) To better understand the long-term effects of the space environment on representative bacteria. The task addresses multiple critical questions in the HSR&T roadmap. This includes needs for methodologies to detect and monitor microbial populations relating to human disease, biofouling of equipment, monitoring performance and integrity of life support systems. This includes spacecraft air, water and surfaces.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	This task addresses a number of critical questions in the general area of address of microbial detection, monitoring and quantification. Each of these questions is generated from a slightly different mission need or concern and they are in fact subtly different. Hence, multiple approaches are being investigated. The results to date and future plans are discussed for the various critical questions in the text below. (1) Critical questions 9c and 10a: #9c- What diagnostic, environmental monitoring, or laboratory technologies need to be developed for the identification of pathogenic microorganisms, and prevention or diagnosis of infectious diseases while in space (e.g., bacterial, viral, or fungal typing in real-time)? 10a - What technologies will monitor, detect, quantify or identify microorganisms that pose a threat to human health during a mission as a countermeasure for preventing further contamination or disease (e.g., bacterial, viral, or fungal typing in real-time)? These questions pose the need for immediate monitoring and detection of specific organisms of known concern. Culture based methods can partially address these needs as well as some of the needs of the related critical questions. The major advantage of culture based technology is that following collection of an air or water sample the collected material is simply added to various culture media without need for further processing. The disadvantage is that not all organisms grow on culture media and organisms of concern might be missed. The alternative is any of several molecular methods that can reliably detect any specific target organism. However, the down side of all such methods is that, at the least, the cells must be broken to release contents followed by purification. In addition, many of the candidate approaches also include a nucleic acid amplification step that then might require development of a specialized instrument. We are exploring the potential of an especially promising approach in which probes that carry a fluorescent label target specific sequences in the ribosomal RNA molecules of the organisms of interest (virus identification will require targeting of genomic nucleic acids). When the probe finds its target, light is emitted at a specific wave length that is then detected. As with other molecular methods, beacons can target any organism of interest. The key advantage of molecular beacons is that they can find their target in a complex mixture thereby in principle greatly simplifying sample processing. Thus, following a minimal RNA purification step the sample would be distributed to individual prefilled containers containing the appropriate buffer and beacons corresponding to the organisms of interest (i.e. different tube for each beacon with some multiplexing possible) much as is envisioned with the culture based assays. The key limitation is sensitivity due to the lack of a nucleic acid amplification step. The choice of rRNA as the target molecule partially solves the sensitivity problem as these molecules are present in thousands of copies per cell. In order to further enhance the signal we are doing three things: (1) taking advantage of newly published data on accessibility of potential target sequences in the ribosomal RNA as well as atomic resolution structural data to design better probe targets, (2) testing beacon fluors that are claimed by their manufacturers to produce much stronger signals and (3) testing a novel “beacon swarm” approach in which we will use multiple beacons targeting different regions of the RNA with the result that the presence of a single RNA molecule will produce multiple flashes at the same wavelength. It is our expectation that these improvements will allow a 2-3 order of magnitude increase in signal with the result that the target sensitivity of being able to detect less than 100 cells will be obtainable. During the past year we completed the design of new beacon targets and testing is now beginning and will be a major thrust in year 2 now that the student working on this area of the project is largely done with classes and other commitments. (2) Critical questions 32g and 134e: #32g and 34e What technologies can detect both anticipated and unanticipated species and events? The key distinction here is that the technology should be able to detect /identify an organism whether its presence has been anticipated or not. This implies a non-routine but very powerful monitoring system. On the Earth, this problem can be addressed in a laboratory setting by sequencing the 16S or 18S rRNA of the organism and comparing the answer to a data base. If the organism has been encountered before, an exact match will occur. If not, a cluster/phylogetic analysis carried out by comparing the unknown organism’s sequence to all known sequences will allow the determination of the organism’s closest relatives or “genetic affinity.” Two approaches to accomplish the equivalent solution without sequencing are under consideration. The first is array hybridization with “signature” probes. It is not sufficient to find a probe for every known organism and put it on a chip as then if an unknown organism is present, none of the probes will detect it. The solution is to instead use a set of probes that includes probes that instead of being unique carry information about relationships, e.g. are “signatures” for various groupings of organisms. We previously showed that large numbers of such signature probes exist. During the past year we developed computational algorithms that allow us to find them. Ongoing computational work seeks to find a minimum set of signature probes (approximately 1000) such that any organism would light up at least 30 probes with every organism having a unique pattern. Related organisms will have a similar pattern. Thus, identification would not rely on a single probe. We also began testing arrays of such signature probes. Unfortunately, the technology we were using was taken off the market and we were thus delayed by having to retool our studies with an alternative technology. This has been accomplished and experimental studies to validate alternative designs of signature probe arrays will be undertaken in the coming year. The second approach which we believe is in fact more promising is the use of mass spectrometry. During the past year we extended our theoretical studies to better understand what might be possible using this technology. The key idea behind the use of mass spectrometry to address these critical questions is that a base specific enzymatic digestion of a 16S rRNA produces a set of fragments that is a reflection of the complete sequence. These fragments can be separated by mass with MALDI-TOF mass spectrometry. A comparison of the sequences of every member of a set of fragments produced with what is expected from known organisms whose 16S rRNA has already been sequenced (over 60,000 strains) allows identification of the organism if it has been previously sequenced. If not, as with complete sequencing, the similarity of

	the pattern (which reflects the underlying sequence) to other patterns will allow determination of genetic affinity. The MS approach would not allow determination of the sequence of the individual fragments but rather only their mass. Because fragments with different sequences but the same composition have the same mass there is substantial information loss. The key finding of the current year's work was that any fragment of length 9 or more would have a mass that is sufficiently unique to be useful. A "catalog" of all the masses of fragments containing at least nine residues will uniquely characterize an organism. In addition, we showed that simultaneous occurrence of 2-3 masses is sufficient to identify many organisms in even very complex mixtures. The use of mass spectrometry is also attractive from an instrument point of view in that a prototype instrument has already been developed and such an instrument would be suitable for use in a variety of medical tests. However, several problems remain. Foremost among these is that cytosine and uracil have very similar masses and hence would be extremely difficult to distinguish with a modest resolution instrument. During the past year we have obtained a solution to this problem. We will first convert the mixture of 16S rRNAs in the sample to DNA templates by amplification and then use a process known as T7 runoff transcription to synthesize a highly purified RNA in sufficient amounts to obtain a mass spectrum. We have demonstrated in doing this that it is possible to replace the uracil residues with another base, e.g. deoxyuracil that has a different mass thereby removing the resolution problem. As a byproduct of this process the RNA can now be uniquely cleaved at cytosine residues by the enzyme RNase A that would normally also cleave at uracil. As a result, it is possible to obtain many additional unique masses by running separate spectra for RNAse T1 and RNase A digestions. During the coming year we will continue explore alternative methods of making the C/U distinction and begin laboratory studies using the complete procedure. THeis will be done initially with single component systems and extended to mixtures as becomes appropriate. Once it is demonstrated that the approach is realistic we will focus efforts on reducing or simplifying the considerable amount of sample processing that would be required. (3) Critical questions 32a and 34a: #32a and 34a: What technologies can be used to detect slow, gradual changes in the chemical and microbial (surface-#35a) environment? The two most promising technologies for this purpose are molecular beacons and mass spectrometry

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